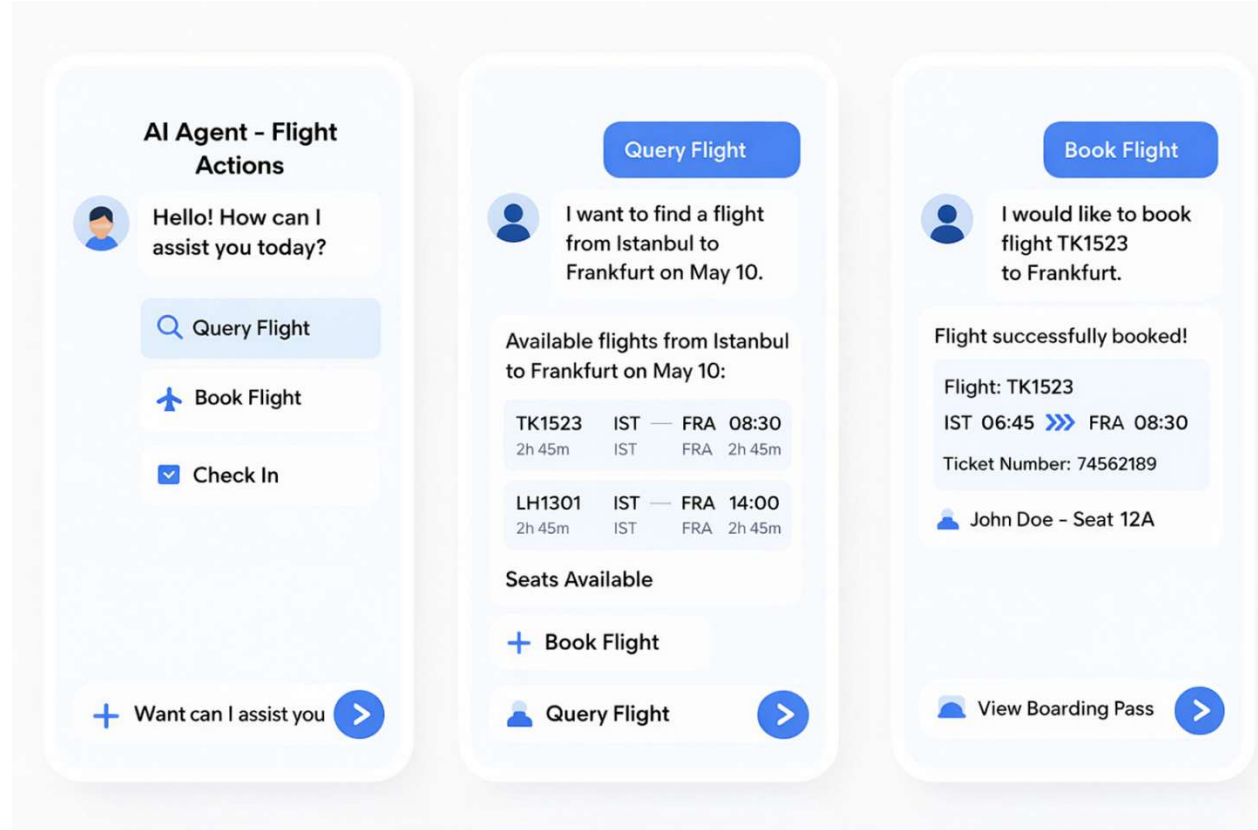


SE 4458 – ASSIGNMENT 2

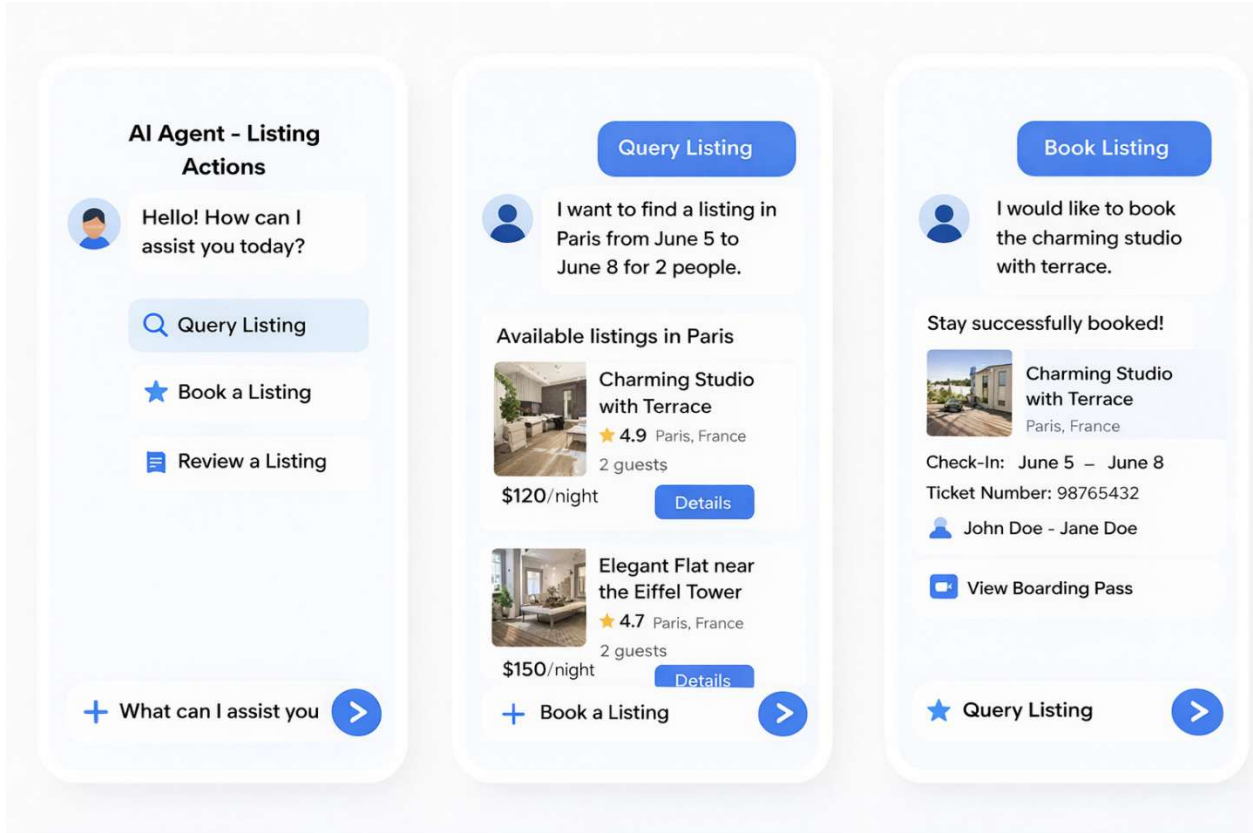
1. Create an AI Agent chat application for the Query Flight, Book Flight and Check in APIs you created in Midterm



Students

Toprak Orman Ilayda Gun EGEMEN ÜNER EGE SARP ERDEM BARIŞ HANSU YAGIZ YUNGUL BERK ATEŞ BATUHAN SALCAN MUSTAFA BERKAY DÜZENLİ	BATIKAN AKDENİZ İDİL BALANDI AYCAN KURT AYFERNAZ BAYGIN EREN KARCI DEMİR DEMİRDÖĞEN SILA BARIŞIK CENK SERBEST
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2. Create an AI Agent chat application for the Query Listing, Book a Listing and Review a Listing APIs you created in Midterm



Students

BAŞAK NUR KORKUT CANSU GÜN İPEK DEMİREL SÜMEYYE ŞENCAN İSMET METE BAYKASOĞLU AYŞENUR TOMBUL ÖZGÜR CAN GÜNGÖR	SENA ASLI TÜRK BEGÜM BAL EMRE AKAR MELİS BABACAN BEYZA TÜRK ÖMER FARUK AKKUŞ DİLARA KUYAR BERKER VERGİ
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EXPECTED ARCHITECTURE

For chat application, use a web frontend framework like React, Flutter, Angular or a mobile frontend like React-native. Flutter.

- Make sure all your API calls go through the gateway
- Consider using Firestore or Realtime Database for Real Time Messaging. See below

architecture recommendation

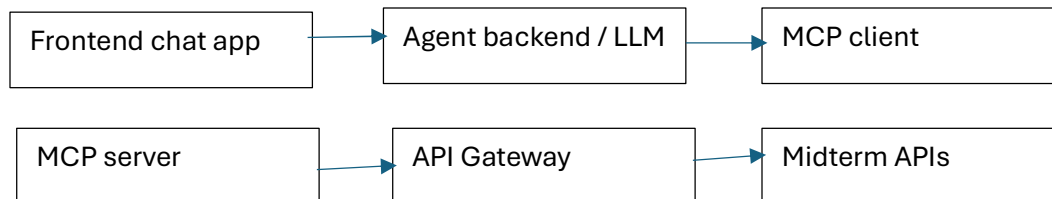
- Call a Firestore cloud function or another API to Call LLM API (OpenAI or local model like Mistral, Ollama) to parse intent and parameters

See examples at <https://github.com/southriver/SE4458-AI-Agent>

- Develop an MCP server for your APIs
- LLM decides which MCP tool to call.
- MCP server maps that tool to the correct gateway endpoint.
- Gateway routes to your midterm API.
- Call Midterm APIs per message text. Assume your chat application uses constant

userid/password for authentication when needed. You can add more APIs if you wish

- Refresh chat API per API responses



You can also utilize websockets with a backend WebSocket Server or Server Send Events with required libraries

Simple implementation idea

Components

- **Frontend:** React or Flutter chat UI
- **Agent backend:** Node.js / Python / Java
- **LLM:** OpenAI, Ollama, Mistral, etc.
- **MCP server:** Node or Python
- **Gateway:** your existing API gateway
- **Midterm APIs:** airline or listing services

Minimal backend logic

- receive chat message
- send message + conversation history to LLM
- allow LLM to call MCP tools
- tool calls hit gateway endpoints
- return response to frontend

DELIVERABLES

- Project does NOT need to be deployed to a cloud app service if you are using a local LLM
- Link to your github code
 - A readme document in your github code repo that has
 - code link to source code of the project i.e github, bitbucket
 - your design, assumptions, and issues you encountered.
 - Include a link to a short video presenting your project (hosted on one drive, google drive, youtube)

Example	
Code	https://github.com/southriver/apiNode
Deployment	https://mpurfkzikk.eu-central-1.awsapprunner.com/api-docs/